

The Natural Philosophy of Universal Matter

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Author's Note

The Natural Philosophy of Universal Matter is a philosophical work and, in relation to physics, should be considered a work of fiction. The reason is that many of its tenets and concepts cannot now—and may never—be directly observed, measured, examined, or researched. The primary reason is that universal matter, described below, is at or below the Planck scale.

Readers may be interested to know that current state-of-the-art scientific technologies can observe objects at about half the Planck scale. At this resolution, science can explore very small objects, such as protons and neutrons. For the particles discussed below, however, scientists would need to halve the resolution again. This would permit observations to within 75% of the Planck scale.

I was a professional programmer and am now retired. My first project used the brand-new IBM PC to gather policy information into fields and then print the many forms needed to submit an application to the home office. For a person with weak typing skills, programming that computer was a godsend.

I mention this because, from the very beginning, I learned that even seemingly simple instructions, such as directing a computer to print a string of text characters, involved thousands of lines of programming code that remained hidden from me. As I moved on to more powerful languages, this fact became clearer.

It is this experience and knowledge that I bring to this philosophy. To this programmer, something as simple as an atom of hydrogen was like that print statement: it held an incredible number of “lines of programming code” hidden from view. The lines of code, of course, are smaller particles of matter held inside protons, neutrons, and the like.

Editorial note: I used ChatGPT 5.6 Sol Medium to review the work for spelling and grammar, ensure consistent punctuation, and suggest the clearest wording for a general audience. Here is the prompt I used to complete these tasks:

Process the refreshed LaTeX file “filename.tex” as a professional editor and proofreader using *The Chicago Manual of Style*, and write a list of errors with their line numbers to output.txt. Put a line break between errors and format each error so that the text following “Original” and “Corrected” starts in the same column. 1. Be consistent about hyphenation: in one review you say it should be ball-of-cells. I make the change, and in the next review you say ball of cells. 2. Leave series join and quantitative join unhyphenated. 3. Always place either series or quantitative before join. 4. Do not hyphenate triple prime. 5. Check that $\{f\}$ and f are used consistently with their definitions in the LaTeX file.

Part 1

The Natural Philosophy of Universal Matter

1 Preface to the Philosophy

Central Proposal This natural philosophy proposes that very small particles of matter pervade the cosmos and that the processes discussed herein compose these particles into the objects of our visible universe.

Presumptions The philosophy builds from these presumptions:

- These particles of matter approach the Planck scale and are therefore undetectable by human senses and scientific instruments.
- This matter pervades our cosmos as a particle field.
- Processes compose these particles into larger compositions.
- When acting individually or as particle compositions, these particles can affect all objects in the field.
- Caveat: These particles do not exist in the Standard Model of particle physics.

Invisible Matter The idea of invisible matter is not new. From ancient times through the nineteenth century, invisible matter was presented as an aether. Then, in the twentieth century, Einstein's theory of general relativity redefined invisible matter as curved spacetime. Now, in the twenty-first century, we present invisible matter as dark matter and dark energy. In each of these concepts, invisible matter was, or is, thought to directly affect regular matter.

Designated Labels This philosophy's particles and particle field are also invisible and affect regular matter. This philosophy refers to the individual particles of matter as "universal matter." The particle field of universal matter is f . The set of all universal matter particles in the field is designated as $\{f\}$.

2 The Universal Join Process

Particle members of the set $\{f\}$ are irregular in shape and size and are not evenly distributed within the field

f . Perturbations in the field allow members of $\{f\}$ to move and act independently of one another.

Any particle member of the set $\{f\}$, such as X, when in motion, may collide with another particle in the field f , such as Y. The collision produces both a complex result and a simple result. The complex result has two possible outcomes:

- Due to the irregular shapes of particles X and Y, X may collide with Y and remain joined, forming a single composite particle composed of two members of the set $\{f\}$. The collision may also introduce rotation in the resulting composite particle.
- Alternatively, X may collide with Y, separate, and remain as the particles X and Y.

In either outcome, there is an additional simple result:

- The collision may cause small fragments of X or Y, or both, to break off from their parent particles. These fragments become separate sub-particles of universal matter, which will be labeled "sub- f " particles. They retain the properties of their parent particles, and energy is transferred from the parent particles as a result of the collision.

As sub- f particles are formed, they become members of the set $\{f\}$. As such, they occupy space in and around their parent particles, forming a localized influence on neighboring particles and other sub- f particles.

Particles that successfully join together are "composite" particles that have a single dimension of size. Additional collisions allow new particles to join with these particles, resulting in either new composite particles or additions to existing composite particles. The collisions and joining of particles of $\{f\}$, sub- f particles, or composite particles impart a second dimension, rotation, to composite particles due to kinetic energy transferred during interaction.

Note One The Universal Join Process clearly involves particle movement and energetic collisions. This philosophy cannot yet address the phenomena, as the necessary objects and processes have not yet been discussed. Particle movement and energy are treated at length in Part 3.

Note Two The following discussion focuses on a direct build path from individual particles of universal matter to a hydrogen atom. This should not be construed as implying that this path exhausts all possibilities. The particles produced by the processes discussed here represent only a few, critically important examples among the myriad possibilities found in nature.

3 Composite Particles

3.1 General

The growth of a composite particle mirrors embryonic human development. In human growth, the embryo leaves its ball-of-cells stage after the fourth week of development, at which point cells begin to form the detailed anatomy of a baby [1].

In the previous section, we saw that a composite particle starts its primary growth stage when one particle of universal matter collides with another particle and the two remain together as a newly formed composite particle. It then begins a ball-of-particles stage by incorporating additional particles of universal matter and continues its development.

Composite particles grow in size through collisions with other particle members of $\{f\}$ or with other composite particles. Sub- f particles accumulate within and around composite particles as growth proceeds. At this stage, composite particle growth mirrors another organic building process, namely plant growth and branch development. As noted by Nakamasu and Higaki, “Various branch architectures are observed in living organisms including plants” [2]. The philosophy will refer to these branch architectures as “grow and branch” strategies. In the growth levels described below, the designations of “growth” and “branching” indicate the dominant developmental process at each stage.

For clarity and discussion, we set out three levels of composite particle growth:

- **Level 1 (CPL1) Growth.** This is the initial growth phase in the formation of a composite particle. Individual members of $\{f\}$ physically join together, increasing the size of the forming particle.
- **Level 2 (CPL2) Branching.** At this level, CPL1 particles join together. The key to growth at this

second level is the relative sizes and directions of rotation of the colliding CPL1 particles. Two types of growth are observed in CPL2 interactions: series joins and quantitative joins.

- **Level 3 (CPL3) Growth.** Finally, the two types of CPL2 particles join to form CPL3 particles. These CPL3 particles combine series join structures and quantitative join structures into single hybrid particles.

3.2 Level 1 Growth

This level involves existing composite particles joining with other $\{f\}$ particles, including sub- f particles, thereby increasing their size. The particles of $\{f\}$ do not aggregate like a ball of string compacted without space between individual particles. Rather, the structure resembles a bird’s nest of sticks, twigs, and plant matter, or an accumulation of dust and hair on a hard floor. Some existing particles connect and form stable interactions with new particles, but none of the particles in the composite structure are connected to all other particles.

As the CPL1 particle reaches maturity, it begins the branching phase. Collisions shift from interactions with other particles of $\{f\}$ to interactions between mature CPL1 particles.

3.3 Level 2 Branching and Development

As a CPL2 particle begins to form, its development branches into two types of growth: series joins and quantitative joins. Given two CPL1 particles, A and B, such that A collides with B, two factors determine whether the result is a series join or a quantitative join:

- **Size**
 - Series joins are most effective when A and B are of similar size.
 - Quantitative joins occur when one particle, for example, A, is larger than B.
- **Direction of Rotation**
 - Series joins occur when A and B rotate in opposite directions.
 - Quantitative joins occur when A and B rotate in the same direction.

The following discussion examines particle rotation as it relates to the point of contact, or collision, between two CPL1 particles. The directions of rotation are not absolute but are relative to the other particle at the moment of collision. At the point of contact, the particles may rotate in opposite directions, or their rotations may be the same.

Figure 1a and Figure 1b illustrate the effect of rotation at the point of contact, labeled “X.” Figure 1a shows that when two particles rotate in opposing directions about their respective centers, they are rotating in the same direction at the point of contact “X.” Figure 1b shows the opposite case. When the rotations, denoted by “R,” are in the same direction about their respective centers, the directions of rotation at the point of contact “X” oppose each other.

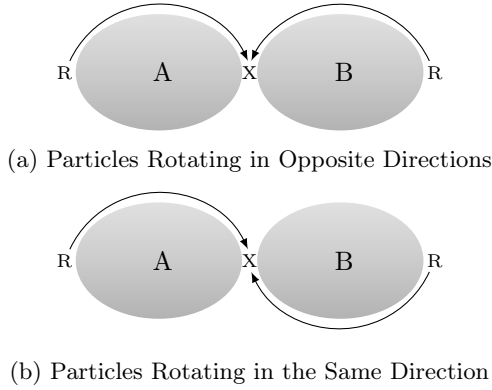


Figure 1: Rotational Directions at Point of Contact

3.3.1 Series Joins

Two CPL1 particles with opposing overall directions of rotation may begin forming a series CPL2 particle. Figure 2 shows that although their overall directions of rotation oppose each other, at the point of contact “X,” both particles rotate in the same direction.

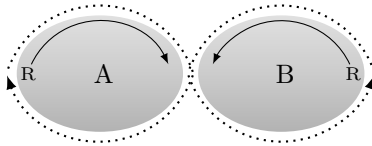


Figure 2: Series Join Particle Exchange

Series joins are most effective when the two CPL1 particles are approximately equal in size. The compatible direction of rotation at the point of contact allows particle A to couple with particle B, and vice versa, thereby forming a persistent series join.

Figure 2 shows that as A and B continue rotating alongside each other, the small particle members of $\{f\}$ in each CPL1 particle, which initially rotate around their respective parent particles, peel off and become entrained in the other particle’s rotation. The exchange of $\{f\}$ particles continues and forms an independent flow jointly held by A and B. This flow forms a figure-eight pattern around A and B, stabilizing the series join.

Additional series joins may occur in a CPL2 particle when other CPL1 particles come into contact with it. As illustrated in Figure 3, the direction of rotation of the newly interacting CPL1 particle, denoted C, must oppose the direction of rotation of CPL2 series member B.

The small surface particles of the newly joined CPL1 particle C peel off and join the continuing figure-eight flow of the existing CPL2 series particle. As a result, the series join evolves into a lengthening figure-eight flow around and between all CPL1 members of the growing chain. In Figure 3, three additional CPL1 particles, D, E, and F, have joined the current members A, B, and C.

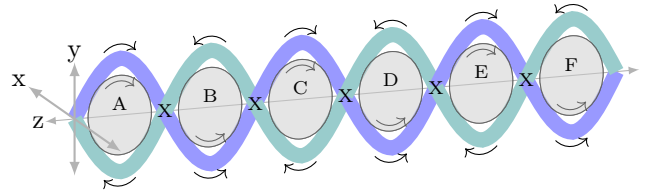


Figure 3: Series Particle Growth

Lattice Growth Figure 4 illustrates that growth may also occur at intersections along an axis. Also, any member of an existing CPL2 series particle may join with another member of the same particle or with another CPL2 series particle. The only requirement is that, at the point of contact, the new CPL1 particle must be fully compatible with all rotational relationships in the series.

Extensive CPL2 series growth at points along an axis of rotation allows the matrix to be viewed as a lattice, as in Figure 4. Additional series joins may establish more lattice area, thereby expanding the growing CPL2 particle. Complex lattices may result when a portion of an existing series join comes into contact with another area of itself and successfully completes the join process, or when the edge of one lattice comes into contact with the edge of another lattice of series joins and successfully forms a single, larger CPL2 particle.

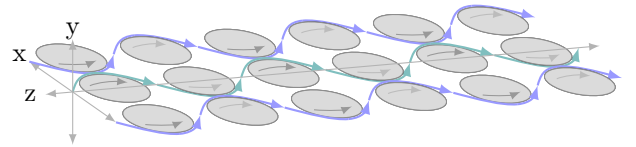


Figure 4: Layered Series Join on the X-Axis

Conductance Complex lattice areas formed by series joins may contain gaps between lattice layers of a CPL2

particle. Then other small, still-forming CPL1 particles may come into proximity to the layered lattice edges of a CPL2 series join. The approaching particle may rotate in either the same direction or the opposite direction relative to the series join and may therefore form either a series join or a quantitative join (see subsection 3.3.2 for information about quantitative joins) with a CPL2 edge particle. The result differs from regular CPL2 joins because of the size difference between the approaching CPL1 particle and the established CPL2 series particle.

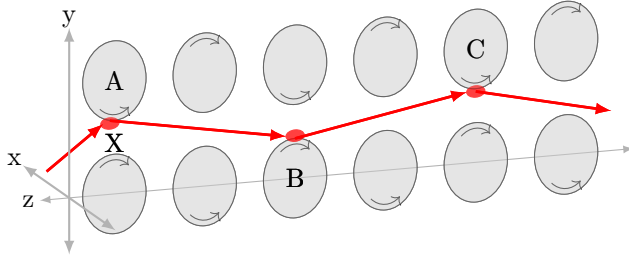


Figure 5: Series Layer Conductance

Figure 5 illustrates a CPL1 particle, “X,” approaching a CPL2 series particle lattice layer. It collides with particle “A,” an existing member of the CPL2, which attempts to join with X based on their respective directions of rotation. The join may be successful; however, the focus here is on a failed join. Following collision, X may rebound inward in the direction of rotation of particle A and continue moving between the surface layers of the CPL2 particle.

Rather than rebounding along a path parallel to neighboring surface layers, X is likely to follow a nonparallel path and collide with another member of the CPL2 series particle. After colliding with A, X rebounds and collides with B, then rebounds in the direction of rotation of B. It subsequently collides with C and rebounds again, eventually exiting the series layer.

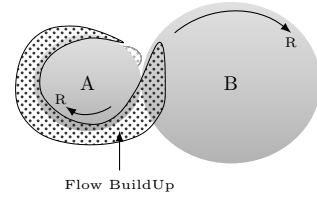
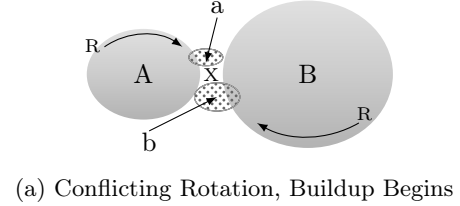
In this manner, X may be conducted through the CPL2 surface-layer gaps and exit at some point within or between the series join surface layers at their far ends. This conducted path may be conceptualized as a flow of matter through the CPL2 layers. Thus, the CPL2 series particle, with its current two dimensions of size and rotation, adds a third dimension to CPL2 growth: that of conductance.

3.3.2 Quantitative Joins

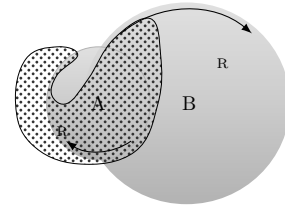
Quantitative joins involve collisions between particles of different sizes rotating in the same direction. Typically, one CPL1 particle, B, is larger than the other, A (Figure 6a). At the moment of contact, A and B ro-

tate in the same direction and begin forming a CPL2 quantitative particle.

Although their overall directions of rotation are the same, at the point of contact “X,” they rotate in opposing directions. The contacting surfaces of A and B begin to accumulate constituent members against each other. A begins piling up matter above the contact point, and B below it (Figure 6b).



(b) B Overflowing A



(c) B Absorbs A

Figure 6: Quantitative Joining

Because of their unequal sizes, the larger particle B accumulates a greater number of constituent members of {f}, eventually overflowing particle A. By overflowing the smaller particle, B draws A into itself and absorbs it (Figure 6c). The two thereby become a single quantitative CPL2 particle.

CPL2 quantitative particles either continue gathering CPL1 particles into themselves or get incorporated into other forming CPL2 quantitative particles. In either case, their size increases.

After quantitative joining, the unified particle continues to rotate. Some overflow particles remain near, but

detached from, the growing particle, forming a cloud of particles moving in the same direction as the primary particle's rotation.

Note CPL2 joining is not always successful. Interacting particles may separate or remain weakly joined, allowing later interactions to break them apart. Collisions may also produce sub-f particles that may later be assimilated in quantitative joins and accumulate in and around forming CPL2 particles.

3.4 Level 3 Growth

Up to this point, CPL2 growth has exhibited two distinct branching strategies: series joins and quantitative joins. Level 3 growth will combine the two: quantitative particles assimilate series particles and become CPL3 particles. The joining process begins as a regular quantitative join, with both the CPL2 quantitative particle and the CPL2 series particle rotating in the same direction. At the point of contact, opposing rotations cause particles of each CPL2 to accumulate against one another.

The lattice structure of the CPL2 series particle slightly alters the joining dynamics. As the larger quantitative particle overflows the series particle, the lattice structure of the series particle wraps around the quantitative particle. The series CPL2 particle is absorbed by the quantitative particle, though not as a single particle. Rather, it is absorbed as a layer on the surface of the quantitative CPL2 particle, adding its conductive properties.

Through this process, CPL3 particles evolve into hybrid particles that combine the volumetric properties of quantitative particles with the conductive lattice patterns of series particles. As CPL3 particles interact with other CPL2 series particles, conductance becomes increasingly significant and acquires importance equal to rotation and size.

4 Advanced Level 3 Growth

As additional CPL2 series particles are incorporated into CPL3 particles, conductive patterns emerge. These patterns arise from the CPL3 particle's ability to conduct matter flow into, through, and out of channels created during the particle's formation. Areas of inflow act as receivers, while areas of outflow act as emitters. These channels play a critical role in further development.

Channel patterns, along with sufficient size and mass, contribute to the sorting of CPL3 particles. Channel

patterns may be classified as simple, effective, ineffective, or complementary. The properties of these advanced CPL3 particle patterns are as follows:

- **Simple:** Some channels successfully conduct particles into, through, and out of the CPL3 particle.
- **Effective:** Many channels successfully conduct particles into, through, and out of the CPL3 particle.
- **Ineffective:** Only a few channels successfully conduct matter into or out of the CPL3 particle.
- **Complementary:** These particles exploit the properties of certain effective CPL3 particles and join at their midsections.

4.1 Simple Conductor

Simple CPL3 particles have a single receiver on one end and a single emitter at the other end. Figure 7 illustrates this architecture.

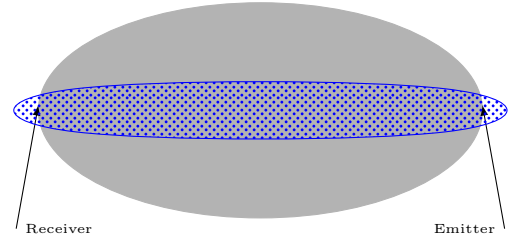


Figure 7: Simple Conductor

4.2 Effective Conductor

Effective CPL3 particles channel particle matter into several receivers or out of several emitters. Each effective particle has two such emitter-receiver circuits, as shown in Figure 8. Conducted particles flow out of an emitter and then flow back into a receiver. In the illustration, the effective particle rotates end over end in the direction labeled R, causing the conducted particles to flow backward, opposite to the direction of rotation, before returning to the receivers.

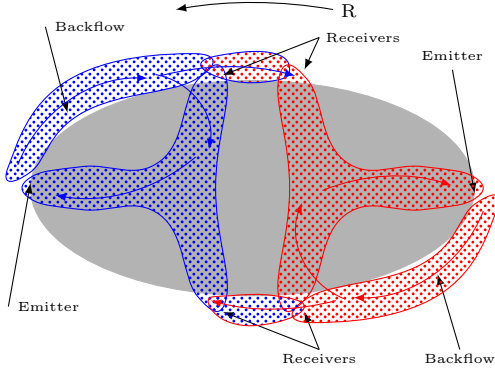


Figure 8: Effective Conductor

Not all conducted matter flowing from an emitter will be channeled back into a receiver. In some cases, particles in the neighborhood of the effective particle may enter the flow through one of the receivers and join existing particles in the loop flowing toward an emitter.

4.3 Ineffective Conductor

Ineffective particles form only a few emitter-receiver circuits. Although areas within these particles can serve as receivers and emitters, they are arranged in a haphazard fashion, creating only a few conducted flow channels. As will be shown in later parts, however, ineffective particles are critical to the formation of regular matter.

4.4 Complementary Conductor

A complementary particle, or “CmP,” is formed when two effective particles join at their midsections in the shape of a plus sign. A representation of a CmP is shown in Figure 9. Here, the CmP is depicted as two shaded ellipsoids. The axes and rotations of the particles are as follows:

- End-over-end in the y - z plane, with rotation about the x -axis. The emitter end is denoted by a red dot.
- Around-and-around in the x - y plane, with rotation about the z -axis. The emitter end is denoted by a blue dot.

Figure 9 may be viewed as two clock faces aligned in three dimensions. The two clock faces bisect each other at $(x, y, z) = (0, 0, 0)$. When viewed along the x -axis, the red effective particle’s clock face appears in the y - z plane, with the red dot at the 12 o’clock position. When viewed along the z -axis, the blue effective particle’s clock face appears in the x - y plane, and the blue dot is likewise at the 12 o’clock position.

The rotation of each particle is relative to the other. For example, if the red particle rotates clockwise, the blue particle will also rotate clockwise, and vice versa. Clockwise rotation is no more or less effective than counterclockwise rotation.

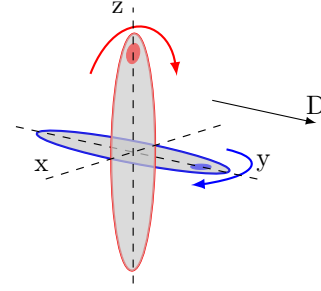


Figure 9: Complementary Conductor Rotations

Figure 10 illustrates how conducted matter flows through one of the two CmP members. The other CmP member joins the illustrated CmP member at its midsection, either above or below it. This structure is similar to the effective particle described in Figure 8, but contains only a single emitter channel through its center. Note the two illustrated sets of receivers/emitters, one at the top and one at the bottom. As conducted particles flow into or out of the receivers/emitters in the midsection of one CmP member, some of the flow may be routed into or out of the other member’s receivers/emitters. The flow between the members’ respective midsections forms a conductive bond between the two constituent CmP members.

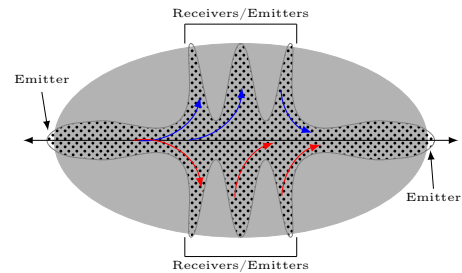


Figure 10: Complementary Conductor Flow

Transitions Summary The CPL1 processes represented a growth stage. CPL2 processes focused on branching out using series joins and quantitative joins. CPL3 processes determined or exploited conductive patterns, setting the stage for the next area of growth: prime particles.

5 Prime Particles

5.1 Primes

The philosophy labels advanced CPL3 particles as “prime” particles. This change in terminology does not indicate any substantive change in composition or properties. Rather, it signifies that prime particles are prepared for a new type of join process.

5.2 Paired Primes

An actively conducting effective prime particle, abbreviated “Ep,” may interact and join with an ineffective prime particle, abbreviated “Ip.” The two primes rotate in the same direction, so the process begins as a quantitative join, and the two particles begin to accumulate constituent particles of {f} against one another in opposition to their respective directions of rotation. However, the additional dimension of conductance modifies the standard quantitative join.

At the point of contact, pools of conducted particles quickly develop. One pool consists of conducted particles from the Ep, and the other consists of conducted particles from the Ip. The labels “R” in Figure 11 denote the directions of relative rotation at the point of contact. Due to the limited channels found in the Ip, the Ep, as illustrated in Figure 11, tries to exploit the Ip’s channel flows.

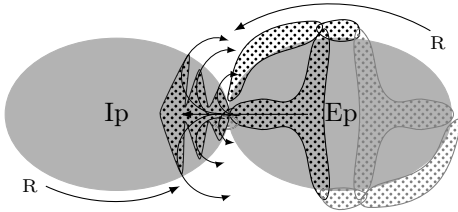


Figure 11: Ep-Ip Paired Prime

Figure 11 illustrates that an Ep and an Ip may successfully join if the Ip possesses a sufficient number of emitter-receiver channels and the Ep is properly positioned to emit particles into the Ip’s receiver channels. The Ep may emit particles into one of the Ip’s receiver channels, shown at the Ip’s 3 o’clock position. The particles then flow into and out of the Ip’s emitter channels located above and below the point of contact. Some of the Ip’s outflow particles may then be channeled into the Ep’s backflow of conducted particles and subsequently flow into the Ep’s receiver circuit at either the 12 or 6 o’clock position, or at both positions. If a sufficient number of these emitter-receiver flow loops form between the two primes, they may persistently join as a single paired

prime particle. Once the join is complete, the paired prime rotates overall in the direction of the Ep.

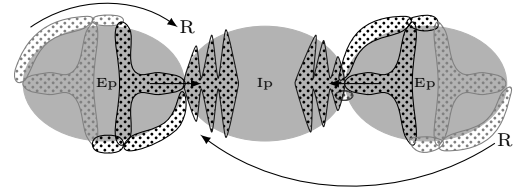
At this stage, the effective and ineffective members of the paired prime are compatible, and there is no rotational acceleration. The stress on conducted particles in a successfully paired prime is nominal and stable.

Success is not guaranteed. An Ep may fail to find a sufficient number of receiver channels in the Ip; indeed, the Ip may have none. Without adequate flow into each particle’s channels, or without sufficient flow occurring in a timely manner, the join effort will fail and the particles will separate. Some transfer of conducted particles may occur, and groups of matter may break off in the collision. The join effort may also produce sub-f particles.

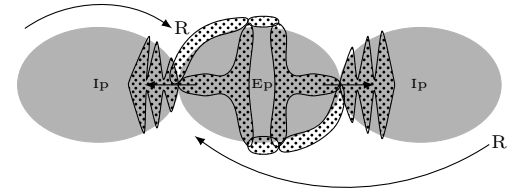
5.3 Triple Primes

Paired primes serve as building blocks that may join with a third prime particle. A third prime needs to join with just one member of the paired prime. The process begins as a quantitative join. A triple prime may form in one of two ways:

- **Active:** The ineffective prime of a paired prime joins with an effective prime (see Figure 12a).
- **Passive:** The effective prime of a paired prime joins with an ineffective prime (see Figure 12b).



(a) EIE Triple Prime



(b) IEI Triple Prime

Figure 12: Triple Primes

Given a paired prime consisting of an effective prime and an ineffective prime, four possible outcomes exist:

- **The Ip of an Ep-Ip pair attempts to join with an Ip: failure.** Weak or nonexistent conducted flows do not establish stable channels.

- **The Ip of an Ep-Ip pair attempts to join with an Ep: success.** (See Figure 12a.) The resulting triple prime is Ep-Ip-Ep, abbreviated “EIE.”
- **The Ep of an Ep-Ip pair attempts to join with an Ep: failure.** Strongly conducted flows oppose and rebuff one another.
- **The Ep of an Ep-Ip pair attempts to join with an Ip: success.** (See Figure 12b.) The resulting triple prime is Ip-Ep-Ip, abbreviated “IEI.”

These joins experience substantial stress while keeping the triple prime particles connected. The stress of an EIE triple join is especially high due to constant opposition between the two effective primes. These two primes, tethered by the middle ineffective prime, may be likened to springs that are continuously compressed, extended, and twisted. The imposed load amounts to stored kinetic energy and contributes to the mass of the triple prime through rotational acceleration.

The CPL1 particles in the conductive joins in the EIE withstand stress because of their construction. No single particle of {f}, nor any subgroup of {f} particles, connects to all constituent particles in a CPL1 particle. The assembly processes described in this philosophy produce loosely connected structures in which constituent particles form coherent larger particles through partial connectivity. Stress on a conductive join exploits this structural property.

6 A Simple Particle Joins a Triple Prime

Both active and passive triple primes may attempt to join with new effective or ineffective primes. Of particular interest is a join between an EIE triple and a simple CPL3 particle, labeled “SCP.” When the SCP contacts the flow shell of an EIE triple prime and both rotate in the same direction, the collision initiates a quantitative join. The EIE triple quickly overflows the SCP. However, the result may differ from the classic scenario in

which the SCP is absorbed, because the nature of the SCP may alter the outcome.

6.1 SCP Join

The SCP tumbles along its x-axis, engulfed by the conducted flow of the EIE. The SCP may then align with the flow direction of the EIE. The receiver end of the SCP begins taking in the conducted flow of the EIE, and the SCP’s emitter end returns that flow into the main stream. The SCP is now anchored in the primary flow of the EIE.

6.2 Quantitative Pull

The conducted flow of EIE matter through the SCP enables the SCP to resist the quantitative pull of the EIE triple particle. Sub-f particles assist in this resistance. These energetic sub-f particles remain in and around the EIE, exerting pressure in all directions. This action serves to:

- Add space within the triple prime, maintaining structural openness and facilitating internal repair.
- Add loft to the triple prime’s conducted flow, preventing it from succumbing to the quantitative pull of the structure as a whole.

The EIE continues its quantitative pull on the SCP. The SCP, anchored in the receiver-emitter flow of the EIE, remains able to resist the pull and spring out and away from the EIE triple prime. Once the SCP successfully joins with the EIE, the pull and spring actions prove to be very stable and persistent.

6.3 Completed Build

Together, the SCP and the EIE form a stable hydrogen atom. The EIE triple particle, with effective prime particles on either side of the ineffective prime particle, forms the proton nucleus. The simple CPL3 particle, embedded within the EIE’s conducted flow, is the electron.

Part 2

Chart of Elements

1 Elements

The process of building the elements does not stop with hydrogen. Let us discuss how the philosophy builds

the remaining elements in the periodic table. While the following discussion focuses on the first three shells of the periodic table, a full presentation with building instructions may be downloaded from an external repository [3].

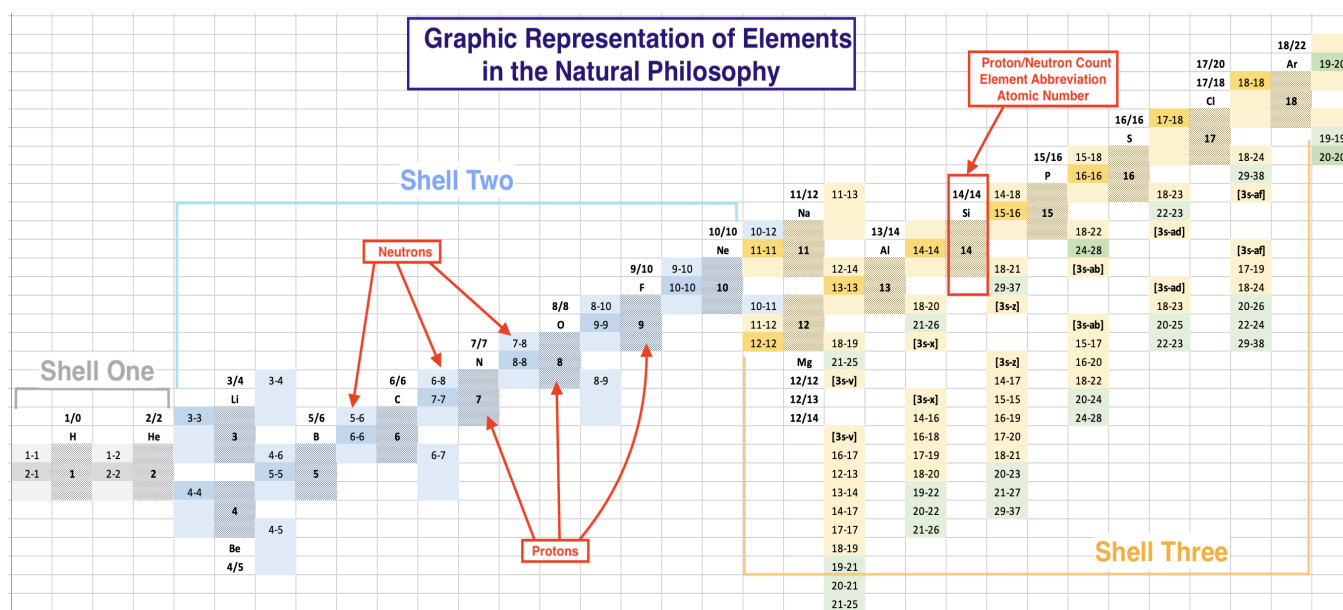


Figure 13: Graphic of Three Shells

Figure 13 illustrates a portion of the Graphics tab in the spreadsheet AtomicStructure.xlsx, which is available in the repository [3]. The first shell, in shades of gray, is located at the far left of the figure. The second shell is in shades of blue, and the third shell is in shades of orange. The proton triplets appear in darker shades of the shells' colors. In each proton triplet, the three cells, read top to bottom, represent an EIE triple prime. Neutrons are always located between the more darkly shaded protons. In each neutron triplet, the three cells, read from top to bottom, represent an IEI triple prime. Protons join only with neutrons.

				3/4	3-4
	1/0		2/2	Li	
	H		He	3	4-6
1-1	1	1-2	2		5-5
2-1				4-4	
				4	4-5
				Be	
				4/5	

Figure 14: Elements Detail

Hydrogen Let us take a closer look at some of the graphical notation found in Figure 14. We will start with hydrogen, shown as a darker block of cells¹ in column

¹As in the cells of a spreadsheet.

two of the illustration and labeled “1.” Above the cells is the letter “H,” hydrogen’s chemical symbol. Above that is the proton/neutron count of “1/0,” showing that hydrogen has only one proton and zero neutrons.

In the lighter shade of gray, just to the left and right of hydrogen’s proton are neutrons. One neutron to its left, in a lighter shade of gray, bears the label “1-1.” This signifies hydrogen’s first isotope, deuterium, which is made up of one proton and one neutron. The first neutron, “1-1,” forms a full “block join” with the proton of hydrogen. A full block join occurs when the three primes of a neutron form a one-to-one join with each of a proton’s three primes. A second neutron, to the right of the proton, forms a full block join with hydrogen’s proton. The neutron, also in a lighter shade of gray, is labeled “1-2” and signifies the hydrogen isotope tritium, which is made up of one proton and two neutrons.

Helium In the fourth column, in a darker set of cells labeled “2,” is helium. Its chemical symbol, “He,” is above those cells, followed by “2/2,” showing that helium is made up of two protons and two neutrons. As we move left to right across Figure 13, we are building up elements, so helium includes hydrogen and its two neutrons. The new designations for the neutrons are “2-1” and “2-2.” The new designations are permanent for the remaining elements, so the cells are given a darker shade to signify that the neutrons now belong to helium, not hydrogen.

Lithium The building of elements from left to right across the illustration follows a “grow and branch” process. We discussed the impact of growing and branching above in section 3. Now, in the fifth column of the illustration, the first shell branches upward to create the second shell shown in shades of blue. To support lithium’s third proton, we need to add a neutron between helium and lithium’s proton. This neutron is found in the three cells located to the left of the darker three cells labeled “3,” and it is labeled “3-3.”

Neutron 3-3 does not form a full block join with the triple primes of helium’s proton. Instead, lithium’s first neutron joins its bottom (ineffective) prime to the top (effective) prime of helium. This forms a “corner join” between the top or bottom prime of a proton and the opposing top or bottom prime of a neutron. The darker shade of the cell labeled 3-3 indicates that its place in the branch chain of elements is permanent.

Lithium needs one more neutron. It is shown above and to the right of lithium’s proton, forms a corner join,

and is labeled “3-4.” Above lithium’s chemical symbol “Li,” the chart shows that lithium is composed of three protons and four neutrons. The fourth neutron, labeled “3-4,” is not connected to any proton further along its branch, and so it is left in the regular shade of blue.

Beryllium As lithium branches upward, the fourth element branches downward. Here, beryllium’s first neutron forms a corner join with helium and is labeled “4-4.” This neutron then forms a full block join with beryllium’s proton, labeled “4.” Below the chemical symbol for beryllium, “Be,” we note that this element has four protons and five neutrons. The fifth neutron, “4-5,” moves downward and forms a corner join with beryllium’s proton. There are no further connections along this branch, and so the neutron remains in the regular shade of blue.

Beryllium has a common isotope that adds a sixth neutron labeled “4-6.” This neutron forms corner joins with the protons of both lithium and beryllium. Note that, in this block of three cells, the cell labeled “4-6” is at the top of the block, in the light shade of blue, while the middle cell is labeled “5-5” and is a darker shade of blue. This indicates that the neutron labeled “4-6” is not permanent in the branch, while the neutron labeled “5-5” is permanent as part of the fifth element, boron (to the right but not in view).

Element Rotation Finally, we must discuss the rotational relationship between elements in this chart. In Part 1, we noted that one of the dimensions for all particle build levels is rotation. Our chart starts with hydrogen, and we must bear in mind that the hydrogen proton rotates end over end about the x-axis of the page. As neutrons and then protons are added to the branch chain, their additional rotations are incorporated into the system. The rotations of the triple primes cause the chain to rotate and loop back on itself, forming spiral layers as the atoms increase in complexity.

Even adding just the first neutron to hydrogen demonstrates, through the resulting rotational complexity, the difficulty of growing and branching in the chart. Not only must the triple prime of the neutron align with the proton’s triple prime, but the neutron must also match the rotational velocity of the proton. Subsequent elemental joins are even more difficult, requiring pressure that is strong and persistent. Therefore, without additional assistance,² only hydrogen and helium, along with trace amounts of lithium and beryllium, are formed.

²Such as at the end of a star’s lifetime or in a laboratory.

Part 3

Particle Movement and Energy

1 Distribution of Matter Within the Field

In Part 1, the philosophy presented the Universal Join Process and discussed how additional processes resulted in building an atom of hydrogen. To facilitate the discussion, we will characterize areas of the field f as either low or high in Universal Join Process activity. Low activity corresponds to a distribution of universal matter characterized as “standard.” Conversely, areas of high activity correspond to a distribution of universal matter characterized as “composite.”

In addition to Universal Join Process activity, standard and composite distributions of universal matter may be characterized by the distance between any particle of $\{f\}$ and its neighbors and by the volumetric density of f field matter. Let us establish, at an arbitrary location within a region of standard distribution in the field matter, a three-dimensional coordinate system with x-, y-, and z-axes. We will consider a single particle, designated “P,” located at the origin of these axes. We will also consider the volumetric density of matter within that distribution.

1.1 Standard Distribution

In this distribution, we consider the distance from any particle of $\{f\}$, such as P_x , to each of its neighbors. We find that the average distance from P_x to any one of its neighbors is approximately equal across all such measurements. If we replace P_x with any other particle of $\{f\}$, such as P_y , and measure the distances to its neighbors, we find that each measured distance and the average of those distances are essentially equal to the corresponding distance and average for P_x and its original neighbors. Thus, we find that the standard distribution of $\{f\}$ matter is homogeneous with respect to the average distance between particles.

We also note that the volumetric density of matter throughout this distribution is roughly uniform. Whether we consider the region of P_1 or the region of P_2 , we see very little difference in volumetric density of $\{f\}$ matter.

1.2 Composite Distribution

The joining behavior of composite particles in the field f is detailed in Part 1. In certain regions, given the vast amount of $\{f\}$ matter, composite joining proceeds extensively, forming large aggregates of composite particles. Significant quantities of atoms and molecules emerge, which may further assemble into larger structures ranging from space dust to asteroids, from small planets to large gas giants, and from stars to galaxies.

Let us consider a star and place P_x at its center on the x, y, and z axes. Again, we measure the distance between P_x and its neighbors. We then relocate P_x outward along one of the axes to a new position, P_{x+1} . Measuring again, we observe results that differ from those of a standard distribution: the average distance from P_{x+1} to its neighbors is greater than the average distance initially measured for P_x . As we continue moving outward along the same axis, repeating the measurements at positions such as P_{x+n} , the average distances to neighboring particles continue to increase relative to those at P_x . As the distance from the origin increases, these measured distances gradually approach those characteristic of a standard distribution.

In this area of composite distribution, the overall volumetric density of $\{f\}$ matter is substantially greater than that in a region of standard distribution. The movement of energy and composite particles is likely the source of this disparity.

1.3 Combined Distributions

This framework suggests that particles of the field f and conventional astronomical objects are related. The observable universe is composed of universal matter, and the relationship in quantity is proportional: regions containing greater concentrations of astronomical objects correspond to regions with higher concentrations of universal matter.

Major compositional objects, such as Earth, exhibit a dense, though not solid, concentration of universal matter at their cores. As one moves outward from the center of an astronomical object, the concentration of field matter decreases, approaching levels found in regions of

standard distribution. This gradient, extending from dense composite regions to more diffuse standard regions, is a fundamental feature of field distribution and plays an important role in particle movement within the field.

Together, these two distribution models of $\{f\}$ particle matter form the first of two foundational concepts necessary for understanding particle movement in the field. The second concept concerns the movement of particles itself: how movement arises and what its source may be.

2 Energy

2.1 Basis

The philosophy presents several examples of how particle conductance and rotation develop:

- When sub- f particles of $\{f\}$ break off from parent particles, they retain the momentum imparted by collisions involving their parent $\{f\}$ particles;
- When additional particles of $\{f\}$ join CPL1 compositions, the additional collisions impart rotational energy to the developing particle;
- Subsequent levels of joining activity convert collision energy to rotational energy;
- Series joins convert rotational energy into conducted-particle energy, which is then used in joins between effective and ineffective prime particles; and
- Quantitative joins convert energy into the small particles used in the join, some of which remain with the larger particle as an accumulating cloud of conducted particles.

From this, we see that composite particles gather, store, and use energy from the collisions that cause and build them. In the philosophy, energy:

- Can be held in rotation around a composite particle;
- Can be found in a particle's series join energy interfaces; or
- Can be transferred to a new composite particle.

2.2 Basis of Energy

Energy is a wave representing the physical movement of a nonphysical body through the field f . This wave may or may not include a particle. If it is included, the particle is not directly part of the wave; it is carried by the energy of the wave and moves through the field. The movement of energy is accomplished through a series of rotations of field matter. Energy moves in the direction of travel, establishing new rotations via quantitative join behaviors at the front of the flow of energy.

As the energy flow moves through the field, it leaves the old rotations to fold back into ambient field conditions. Energy moves in the direction of travel, but field particles do not move forward except through rotation.

The movement of energy is not free. Some of the energy is expended in spinning up new rotations of field matter. At the field level, energy is transferred from one rotation to the next. The transfer of energy is extremely efficient, but it is not perfect. Some of the particles held in energy rotations fail to spin up new field matter and fold back into the field. Regardless of the energy wave's breadth and depth, it diminishes over time, inevitably coming to an end. Its energy will dissipate and the affected particles will settle back into the field.

Let's consider an event, perhaps the explosion of a star, that imparts energy to the field. The event, though perceived in our observations as a single large event, is not a single expression of energy at the field level. Rather, the energy is expressed as a number of much smaller, localized events at the field level. The number of local events correlates with the breadth and depth of the predecessor event's total energy. Let us detail one of these local events.

Each local event comprises a series of rotations that vary in size and form an elongated elliptical shape, as illustrated in Figure 15. As the event moves in the direction of the energy flow, it converts its potential energy into quantitative join efforts to build rotations. As the energy of the event moves forward, the energy rotations in its wake decay and lose their energy.

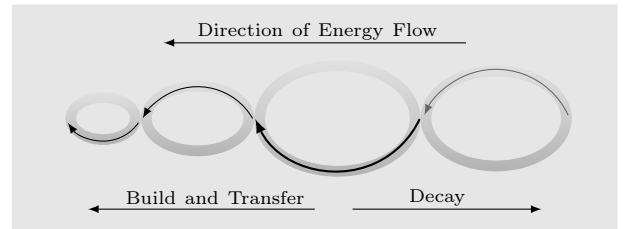


Figure 15: Energy Rotations Moving Through the Field

Now, let us consider an energetic event that includes a physical body as well as energy.

2.3 Composite Particle Structure

Energy, as noted in Figure 15, moves as a group of local rotations of field f matter. When a composite particle expresses energy, there is a correlation between its constituent particles and localized rotational events in the field. Here, the group is expressed by a subgroup of particles, each of which corresponds to a local rotation event. Let's take a closer look at a typical composite particle to better understand this relationship.

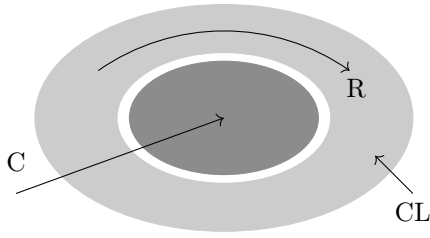


Figure 16: Composite Particle with Its Conducted Shell

Figure 16 illustrates the core (C) of a composite particle and the matter in its rotating conducted-particle layer (CL). The direction of rotation (R) in the conducted-particle layer (clockwise in this illustration) is arbitrary.

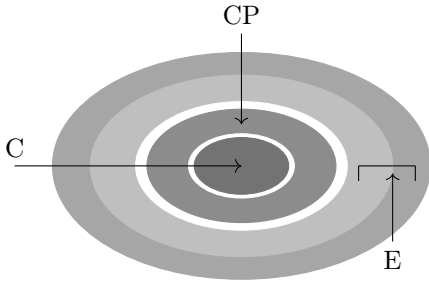


Figure 17: Adding the Energy Shell

Figure 17 adds the energy particle layers, labeled E, to the particle. They rotate in direction R, the same direction indicated in Figure 16. The energy layer has inner and outer sections composed of layers of particles. The rotational velocity of the inner energy layer best matches the velocity of the CL layer. The outer layer acts as an intermediary between the particle and the field f .

The conducted particles held in the E layers are indicative of the total amount of energy held by the composite particle. Because an energetic event affects the local field, however, the energy of the composite particle and that of the energy wave are created simultaneously and integrated.

2.4 Integrated Energy and Particle

As it moves forward in direction D, the composite particle will attempt to find its place within the primary rotations as illustrated in Figure 18. The outer energy shell, rotating as indicated by the arrow, interacts with particles rotating in the same direction. The matter held in the rotations must have mass equal to or greater than that of the composite particle to “pick up” the particle. If this is true, the energy wave and the composite particle will be integrated into one combined object.

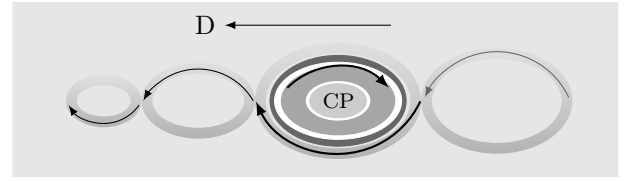


Figure 18: Integrated Energy and Particle

The amount of conducted matter in these forms indicates the amount of energy available to the combined energy wave and particle.

The event does not always produce a successful combination of the energy wave and particle. Less than 100% of the energy that precipitated the event is transferred to the (new) energy wave/particle. For example, the composite particle may not find a compatible energy wave in which to reside, in which case both components are lost to the field. Alternatively, as an energy wave begins to assemble, its rotations may become disordered, causing the wave to fail and fold back into the field.

3 Particle Collision

Movement is caused by a precipitating event, such as an energetic impulse or a collision between two bodies. Let's focus on the latter and, to aid in the discussion, we'll consider two particles:

- One that will serve as the **predecessor particle** (P_p); and
- A second particle that will play the role of the **successor particle** (P_s).

Now imagine that P_p collides with the stationary successor particle P_s . Let's look at how P_p causes motion in and transfers energy to P_s . To do so, let's redraw Figure 18 and add the successor particle (far left).

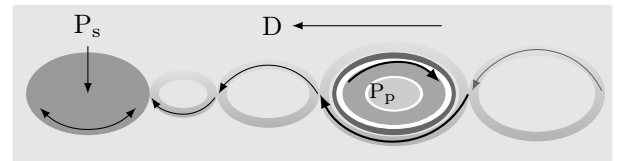


Figure 19: Adding the Successor Particle

The leading energy rotation of P_p first contacts P_s . P_s then begins to “hunt and peck” for the best fit as it progresses through the leading energy rotations of P_p . As it progresses through the energy rotations of P_p , P_s seeks a rotation within P_p 's energy wave that turns in the same direction and is equal to or larger than P_s in size.

This discussion presumes that P_p has sufficient energy to engage and deflect P_s . If not, the two particles will collide, and the resulting activity will subside as both particles settle back into the field. In a successful collision, P_p 's energy wave captures P_s within a rotation that is capable of carrying it.

3.1 An Energy Wave

Figure 20 depicts P_p colliding with P_s .³

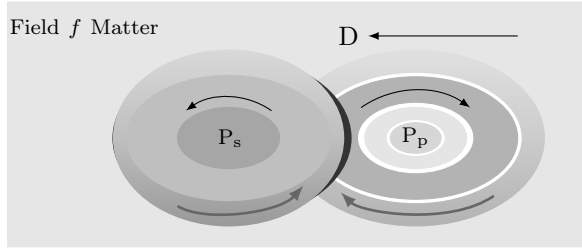


Figure 20: Impact

For the sake of simplicity, we'll presume that the best-fitting predecessor energy rotation for P_s is the one next to P_p . The best-fitting rotation may, however, be one of the preceding energy rotations along the direction of travel.

In either case, P_s is rotating in the direction opposite to P_p 's energy rotation. The rotation forms a wave of rotating matter that presses against P_s and, if P_p has enough energy, begins to carry the successor particle in direction D. Two subsequent actions occur:

- The conducted matter flow from P_p rolls against P_s . This flow forms a rolling wave of conducted matter pushing P_s in direction D; and
- P_p is in motion and, in the collision with P_s , forms an indentation in both its outer and inner shells. This action compresses the energy shells of P_p . The compression acts like a spring, adding to the outward push against P_s in direction D.

P_p 's energy wave forms a rolling wave of conducted matter that transfers energy to P_s by initiating and accelerating rotations in the successor particle. P_p also transfers energy to P_s through its spring-like recoil from the collision. As P_s accelerates away from P_p , it brings its new inner shell up to speed with its conducted energy shell and, given enough transferred energy from P_p , begins expressing energy by adding conducted matter from the field to its inner and outer shells and by developing its own energy wave of rolling matter. Thus, due to the combined energetic actions of P_p , P_s springs away in direction D.

³Figure 17 illustrates the configuration details of P_p .

This energetic action also affects P_p . Its spring-energy action pushes back in the opposite direction of D. This may cause P_p to move more slowly in direction D, to stop, or to rebound in the opposite direction. Both particles, however, pay a price and lose energy to the field.

3.2 Movement Through a Standard Distribution of Field Particles

P_p and P_s are examples of particles in motion in the field f . Also, in previous discussions, we found that energy itself can and does move through the field. As noted above, in Figure 18, any such particle or energy wave has an energetic series of rotating field matter preceding it as it moves in any such direction. Let's now call this preceding series of rotating field matter the "interface" of the particle/energy wave as it moves through the field.

While the interface precedes the particle/energy wave and rotates field matter, the rotating field matter does not travel along with the particle/energy wave. The interface is always transferring the energy wave rotation to new field matter. The particle (if there is one) then moves into the space of the interface, leaving the previous rotation to settle behind it.

3.2.1 Huygens's Principle

Huygens's principle states that "every point on a wave front is a source of wavelets that spread out in the forward direction at the same speed as the wave itself" [4]. The particle or energy wave described above is a single instance of one of these wavefronts. The field matter in the interface's leading rotation bows outward, as illustrated in Figure 21, and moves in direction D.

Huygens's principle can be summarized as follows:

If a region has a slower speed of light, it will tend to collect light from the surrounding region. If a light-signal wavefront encounters a light speed gradient across its surface, with a faster natural speed on one side and a slower speed on the other, the retardation of the wavefront's "slower" side will steer the wavefront towards the slower-speed region [5].

Huygens's "light speed gradient" may be equated to the density of the field f . As noted in other discussions, the standard distribution of particles in the field has a homogeneous density, illustrated in Figure 21 as a flat, light gray background. In this scenario, the average distances between the interface and the field are viewed as the darker gray fill between the plotted values for the particle/energy wave and the field.

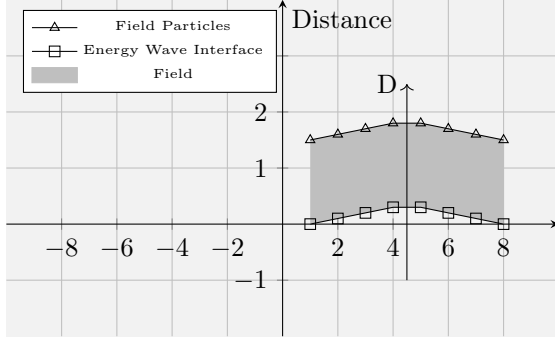


Figure 21: Standard Field Distribution

In Figure 21, the shading between field particles (shown as triangles) and interface particles (shown as squares) represents the distance between corresponding particles. Note that the density and interparticle distance between the interface and the field (illustrated as a darker gray fill) are uniform across the interface. This means that there is an equal ability of particles on the leading edge of the interface (squares) to attempt to join with field particles (triangles), transfer energy to them, and add them to new interface rotations. This results in no change to the path of the particle/wave.

This scenario complies with Huygens's principle. In an area of standard field distribution, there are no regions in which light travels more slowly. Thus, there is no slower side that will steer the wavefront toward the slower-speed region. The energy wave and the optional particle will continue moving in their original direction.

Even for objects moving at subluminal speeds, the effect is the same. Given the homogeneous density of the field, no portion of the particle's leading edge encounters a different drag coefficient, so its path does not change.

In areas with a standard distribution, or density, of field f matter, the interface's rotating matter achieves a nearly one-to-one transfer: each particle in the interface can induce motion in one particle in the field f , gathering the approaching field matter into new energy rotations.

Though the interface exploits regular joining actions to induce rotation in the field, not every field particle is caught in the join; the interface therefore moves on

to another particle. Alternatively, a field particle may become caught but then be lost by the interface; in that case, energy is expended for no gain. Thus, the interface energy may be lost to the field.

This suggests that the density of the oncoming field matter offers some resistance to converting wave/energy to new rotations. The cost, though, is applied equally across the interface and no section of the interface sees more or less resistance. In this situation, the effect on direction is negligible.

3.3 Movement Through a Composite Distribution of Field Particles

This is not the case in areas of composite field distribution. Here in the field f , we find that the density of the field matter changes from its highest level at the core of a massive object to lower levels as we move away from the core. Figure 22 represents an energy wave (with or without a particle) moving in direction D through one of these fields, where the density of field particles is higher on the left.

Here, the density of field matter is higher at the center point (0,0) on the axis grid (shown in darker grays). Density decreases as one moves away from the center (shown in lighter grays). As before, the shading between field particles (shown as triangles) and interface particles (shown as squares) represents the distance between corresponding particles.

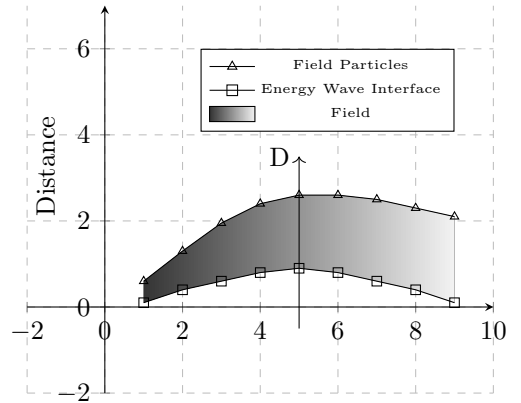


Figure 22: Composite Field Distribution

Moving laterally across the interface, we find that the particle/wave's ability to transfer energy to new field rotations is unequal, with the left side now differing from the right. The figure shows that conditions on the right side of the interface resemble those in a standard distribution; that region is homogeneous and has a negligible effect on the particle's path.

However, results are different on the left side of the interface where the increased field density brings particles closer to those of the corresponding interface particles. Consider the leftmost points in both the plotted triangle and square lines. As interface particles (squares) interact with field particles (triangles), each particle is more likely to gather up more than one field particle. This likely scenario persists for some distance along the interface to the right.

3.3.1 Snell's Law

We now see an inequality of particles gathered up in the new, emerging energy rotation: there are more particles in the energy rotation on the left than on the right. The greater number of field particles on the left side equates to increased density in the field. This difference in density follows Snell's law, affects the object's path, and causes it to refract toward the normal.

Part 4

Gravity

1 Composite Distribution of Field Matter

1.1 Einstein's Curved Space, Time, and Gravity

John Wheeler, in a lecture, tells us: "Matter tells space how to curve. Space tells matter how to move" [6]. We move in a unified field made up of space and time, and these two aspects are unified. As we move through space, our speeds are more regular, but in time, we move at the speed of light. Given general relativity, an object at a greater distance from Earth's surface, such as my head, is moving faster than a closer object, such as my feet. Thus, in terms of time, we are tilted "forward" in time; that is, we fall forward into time.

1.2 Composite Distribution of Matter

This philosophy presents "curved spacetime" as a composite distribution of f field matter. As presented in Part 3, the density of a composite distribution is proportional to the mass of the object and decreases from a higher density in the object's core to lower densities as one moves away from its center into the surrounding space. The center of the composite distribution of field matter and the massive object are the same.

It must be stated clearly that the composite distribution of matter is not caused by the massive object; rather, the massive object results from compositional growth within the composite distribution as described in Part 1. The composite distribution of f field matter and the visible massive object are a single system of matter.

Ethan Siegel indicates that all of the astronomical objects we can perceive account for only about 5% of all matter. The balance of the universe is about 27% dark matter and 68% dark energy [7]. Extrapolation then suggests that when we observe an astronomical object, such as our planet, we are observing only about 5% of the matter that makes up our planet. The remaining matter that constitutes Earth is 27% dark matter and 68% dark energy. The composite distribution of {f} matter that makes up our planet is the sum of these three amounts.

The discussions in the previous parts of this philosophy have shown how universal matter interacts, forming ever larger compositions of field f matter. They have also discussed how the distribution of field matter is the basis of energy and its movement through the field. The discussions in this part will use a composite distribution of matter and the ideas and laws of Huygens, Snell, and Bernoulli to explain the phenomenon of gravity.

1.3 Spacetime Terminology

We might pause for a moment to reflect on Einstein's use of the terms "space" and "time." When his papers on relativity were published, he did not know of galaxies, dark matter, or dark energy. He knew, however, that space and time curved around massive objects, so his use of the terms "space" and "time" best suited the scientific understanding of his era. Today we know of galaxies (as Einstein did before his death) and the dark forms of matter and energy. Let's now put forth a better term for the concept of spacetime.

Using the concepts of this philosophy, we are able to see that a massive object and its curved spacetime are a single system of matter; that is, they form a composite distribution of {f} particle matter. We will abbrevi-

ate this as “CSM,” a composite system of matter. All fully developed astronomical and scientific objects, from atoms to galaxies, are CSMs. This philosophy, though, will discuss only CSMs such as planets, stars, or galaxies. Also, it should be noted that many CSMs are intertwined and interrelated. It is fairly common for a star-system CSM to house several planetary CSMs and for a galaxy to house many star-system CSMs.

1.4 Motion

We have identified composite distributions of $\{f\}$ matter as the source of curved spacetime. We have also combined spacetime and curvature into a single concept: a composite system of matter, a CSM. There is another general idea needed to discuss gravity: that of moving and motion.

We will see that gravity is an effect produced when matter is in motion in or moving through a CSM. Motion is passive while moving is active. Passive motion occurs when objects appear to be motionless, such as when you or I sit in a chair. Active moving occurs when an object, such as a rocket ship, has its own energy and velocity while moving through the field f .

Motion is a subtle property of nature. All objects (that we can know) are in motion. We do not perceive much, if not most, of the motion extant in our world. Our planet rotates on its axis and orbits the Sun; the Sun rotates on its axis; and the solar system rotates around the center of our galaxy. Our galaxy is also in motion (moving toward the Great Attractor). All the things in our various astronomical systems are in motion. Thus, we can never be “still.” Even when an object is passive in nature, its inherent movement satisfies gravity.

It is easy to overlook that an active, moving object is also in motion, so that aspect of the object is unchanged. But if it is moving because of some internal propulsion (such as a rocket) or some energy granted by its inception (such as a comet), it is also actively moving, and this movement affects gravity.

2 Gravity

2.1 Sources

Gravity is not a single action; rather, it is a mixture of three separate sources. The sources are described by Huygens, Snell, and Bernoulli.

2.1.1 Huygens

Huygens’s principle serves to illustrate how waves and particles appear to be attracted to a massive object. In

an article in *LibreTexts Physics*, the writer reminds us that “The law of refraction can be explained by applying Huygens’s principle to a wave front passing from one medium to another. Each wavelet ... was emitted when the wave front crossed the interface between the media. Since the speed of light is smaller in the second medium, the waves do not travel as far in a given time, and the new wave front changes direction (... towards the normal...). This explains why a ray changes direction to become closer to the perpendicular when light slows down” [4].

In Part 3, we correlated a particle/wave’s movement through the field f with a single instance of a Huygens wavelet. Thus, the attraction of an object to a massive object can be better viewed as a refraction of the object toward the normal of the massive object. Keep in mind that in Part 3, the philosophy noted that an apparent solid object is, at the field level, a cloud of concerted particle/waves.

Note Huygens’s ideas are typically applied to light. It does not seem that we should be considering light in this discussion of gravity. However, keep in mind that the philosophy breaks objects down to their particle/wave properties as they are interacting directly with the field f . At this level, the philosophy’s particle/waves are “light,” though they are slowed appropriately by refraction in a CSM.

2.1.2 Snell’s Law

Snell’s law derives from Huygens’s principle. Snell does not change the general ideas of Huygens; rather, he sets out mathematical formulae that one can use to calculate the effects of light moving from one medium to one of greater or lesser density. Now we will add a CSM, motion in and movement through it, to the picture.

We understand that the CSM has a density gradient from its edges to its core. The system of the CSM has a single, common center. The CSM changes position through movement and may also rotate. The density gradient of the field f is always moving as it interacts with a particle/wave whether the particle/wave is in motion or it is moving.

Considering field density in relation to Snell’s law is not a unique idea. Manuel Alfaro states:

Once you let spacetime have a density parameter

$$\alpha(r) = R_s/r$$

and corresponding refractive index,

$$n(r) = 1/\sqrt{1 - \alpha(r)}$$

Snell's Law becomes the inverse square law. (...this...) derivation rests on three Fundamental Density Theory (FDT) primitives:

- Density parameter (outside a mass M):

$$\alpha(r) = R_s/r = 2GM/(c^2 r)$$

- Refractive index of spacetime in that density field:

$$\begin{aligned} n(r) &= 1/\sqrt{1 - \alpha(r)} \\ &= 1/\sqrt{1 - 2GM/(c^2 r)} \end{aligned}$$

- Extended Equivalence Principle (EEP):

$$a = -c^2 \text{grad}(\alpha)$$

You do not add a separate “gravitational force.” Matter just sets $\alpha(r)$; $\alpha(r)$ sets $n(r)$; $n(r)$ tells every trajectory how to bend.

In FDT, gravity = density gradient = refractive index profile [8].

In Alfaro's theory, spacetime has a density. In this philosophy, spacetime is a CSM with a density gradient. With Huygens and Snell, we clearly have a basis for particle/waves curving toward the normal of a CSM. The curvature increases slightly with each moment of motion or moving, but this does not explain the acceleration. How, and why, does a particle accelerate as it curves inward?

2.2 Bernoulli

In Part 1, we discussed Bernoulli in relation to a series join. The discussion in Part 3, specifically subsection 3.1, allowed us to correlate an energy wave with an instance of a Huygens wavelet.

Charles Q. Choi, in his article in *Inside Science*, says “When it comes to ‘quantum gravity’ theories that seek to reconcile quantum mechanics and general relativity, there are currently two main scenarios. One suggests the force of gravity can be described in terms of packets of energy known as gravitons ...The other suggests the fundamental constituents of spacetime essentially condense together like a fluid” [9].

The “other” theory is well suited here as it seems reasonable to suggest that the field f may be viewed as a fluid. In a standard distribution of field matter, the density of the fluid is very low, but not zero. In a CSM, the density of the fluid is much higher and provides a basis for fluid interaction.

In a Bernoulli system, the fluid moves through a tube, for example, with energy e at velocity v . As the fluid encounters an area of resistance in the tube, such as an area with a smaller diameter, e remains constant, but v increases due to conservation of energy. Should the restricted area end and the diameter increase, v will decrease accordingly.

As the particle/wave moves through the CSM, we see from Snell's law that the particle/wave follows a curved path toward the normal, and we view the CSM as a fluid. In Part 3, specifically subsection 3.2 and subsection 3.3, our discussions introduced the particle/wave's interface with the field. The interface was characterized as a curved area at the front of the energy wave, and we have correlated the energy wave and the particle/wave with Huygens wavelets. You may wish to refresh your understanding of the idea by reviewing Figure 21 and Figure 22. We will focus on the latter figure.

Please step back with me and recall that our particle/wave is moving out in front of an energy wave. The goal of the particle/wave is to spin up new field particles and thereby move the energy wave through the field. Now one side of this energy wave is encountering a field that is denser to one side of the interface. Through this effort, the particle/wave is able to spin up more field matter than its interface neighbors.

The particle/wave and the energy wave move along their path through the field with energy e and velocity v . The field on the side of greater density acts as a restriction in the Bernoulli sense, and so v increases on the denser side while e remains constant. The particle/wave continues along a curved path ever closer to the normal. As it does so, the particle/wave's interface rotates until the center of its face points directly toward the greater and increasing density of the CSM. This allows more of the interface particles in front to share in the greater field density and thereby increase the particle/wave's velocity.

The density of the CSM field continues to increase, the energy of the overall energy wave remains constant, and the particle/wave's velocity continues to increase. The increasing density of the CSM field drives the increasing velocity of the particle/wave.

Part 5

Time

1 Introduction

The online *Encyclopedia Britannica* summarizes time as follows: “...measured or measurable period, a continuum that lacks spatial dimensions. Time is of philosophical interest and is also the subject of mathematical and scientific investigation” [10]. Helmenstine writes, “What is time, exactly? Physicists define time as the progression of events from the past to the present into the future. Basically, if a system is unchanging, it is timeless. Time can be considered to be the fourth dimension of reality, used to describe events in three-dimensional space. It is not something we can see, touch, or taste, but we can measure its passage” [11].

Certainly, time is complex for both philosophers and scientists. Although we sense and discuss time, there is no agreement about the nature of time; nevertheless, we cannot deny its existence. The following discussion is not meant to be an exhaustive survey of the philosophical or scientific topic of time. Instead, we will look to this philosophy for insight into the nature of time. Let’s begin the discussion by listing a few areas for consideration.

2 Areas and Types of Time

In these short quotations, we see that time can be measured using a clock. We can also see that time can be used to measure a progression or continuum of events from the past through the present and into the future. Time can also be considered the fourth dimension of spacetime, helping to make sense of changes in three-dimensional space. As Helmenstine suggests, time is change. Related to change are the areas of the arrow of time and spacetime configuration.

Even a brief survey of time-related articles brings up a considerable number of areas related to time. For instance:

- Arrow of Time
- A-theory vs. B-theory
- Second Law of Thermodynamics
- Block Universe
- Science, or Measured Clock Time

- Existential Time
- Changes and Configuration

Let’s set the stage: In the second row back, in the center, is the arrow of time, or “AoT.” Seated in the first row, arranged arbitrarily from our left to our right, are A-theory and B-theory, the second law of thermodynamics, the block universe, Ellis with his evolving block universe, science, existentialism, changes, and configurations.

There are significant discussions in this philosophy that we will relate to time:

- First and foremost, the universal join outlined at the beginning of Part 1;
- The beginning state of the universe, that is, the original standard distribution discussed at the start of Part 3; and
- The CSM (composite distribution of field f matter) discussed in Part 4.

Let’s begin with the AoT, the second law of thermodynamics, and changes.

2.1 AoT, Second Law, and Changes

At the beginning of Part 3, the discussion suggested that, starting with a single instance of universal joining, we could reverse events such as these and find an original point where the field f matter had not experienced even a single universal join. This would present us with a perfect source of a standard distribution of the field matter. No joins, no changes, no time. No composite distributions of field matter. In this light we could say that this standard distribution of field matter was the initial state of entropy. With the first universal join, the universe began its transition of order to disorder.

Many writers suggest, or even flatly state, that the idea of zero or minimal entropy is encapsulated in the second law of thermodynamics and that the law is the source of the AoT. Many see the beginning state of entropy in the singularity that was the source of the Big Bang and in the earliest minutes of the universe. It was then “hot and dense” and had all the things needed to make what we see and explore today.

Eddington (1948, pp. 34–35) primarily associated the AoT with the second law as follows:

Without any mystic appeal to consciousness it is possible to find a direction of time on the four-dimensional map by a study of organization. Let us draw an arrow arbitrarily. If as we follow the arrow we find more and more of the random element in the state of the world, then the arrow is pointed towards the future; if the random element decreases, the arrow points towards the past. That is the only distinction known to physics. This follows at once if our fundamental contention is admitted that the introduction of randomness is the only thing which cannot be undone.

I shall use the phrase “time’s arrow” to express this one-way property of time which has no analogue in space. It is a singularly interesting property from a philosophical standpoint. We must note that

- (1) It is vividly recognized by consciousness.
- (2) It is equally insisted on by our reasoning faculty, which tells us that a reversal of the arrow would render the external world nonsensical.
- (3) It makes no appearance in physical science except in the study of organization of a number of individuals. Here the arrow indicates the direction of progressive increase of the random element [12].

Eddington’s use of the words “random elements” is interesting. We can suggest that random elements are somewhat analogous to the “ordered” and “disordered” precepts of the second law. The second law is well tested in small, measurable environments and at the cosmic scale. As Gonzales-Espinoza and Pevón note, the entropy of the universe is “overwhelmingly contributed by the entropy of the cosmic horizon (about 10^{122} times the Boltzmann constant). Supermassive black holes and the cosmic microwave radiation, alongside the cosmic sea of neutrinos, come behind by 18 and 33 orders of magnitude, respectively” [13].

It is harder to observe and measure random elements. Eddington suggests that they are found in the “study of organization” and notes them as “...the only distinction known to physics” [12]. Unfortunately, Eddington’s “time’s arrow” connection between the second law and the AoT is careful but indirect. We are left with no direct physical source of the second law or the AoT.

This is not the case in this philosophy, which shows that sub-f particles, the byproducts of the Universal Join Process discussed in Part 1, provide a direct physical link between universal matter, the second law, and the AoT: Whenever and wherever universal joins occur, or whenever any type of matter interacts, sub-f particles of matter may be produced. Universal joins occur most frequently in developing CSMs. However, in a mature CSM, the interaction of matter is of greater interest.

We live in a single-use universe that complies with the laws of conservation. Since the beginning of the universe, matter in any form has not been created or destroyed. But the philosophy does show that interactions change matter from one form, universal matter, to another: sub-f particles of matter. The changes are one-way only: interactions cause universal matter to break down into smaller particles of energetic matter. There is no process that brings the small particles back together into full particles of universal matter.

We “feel” the slow, deliberate transition of regular particles of universal matter to sub-f particles as our “sense” of change. Whether discussed as the second law of thermodynamics or as the arrow of time, change is, at heart, caused by the universe’s transition from regular {f} matter to sub-f particle matter. We can discuss sub-f particles of {f} matter as the “dust” of constructing CSMs or as “rust” that seeps in and breaks down CSMs.

2.2 Configuration

We can speak of the relationship between sub-f particles of {f} matter and the AoT in a more generalized sense. We can consider any state of sub-f particle composition and suggest that it is part of a more general state: that of the CSM. In this way, we can speak of both the state of sub-f particle matter and the state of the CSM by referring to their “configurations.”

There is another way that we “sense” changes: through the various configurations of spacetime in our greater or lesser CSMs. Let us consider the configurations of our local CSMs and the nature of “now.”

The CSM of Earth and the CSM of the Sun are ever-changing in their rotations, motions, and locations. Please keep in mind that our CSM world is also moving in deeper ways such as rotating with our galaxy. All of these motions and locations may be considered a series of “nows” and locations. Each now frame in the series represents a specific configuration of CSM matter. Because of motion and changes in location, any specific CSM configuration occurs only once.

The configurations of our now frames depend upon and relate to all the now frames of all CSMs and standard distributions in the universe. We cannot hold our now frame as independent; to reconstruct it, one would have to reconstruct all the related now frames in the universe.

We sense “time” in the ever-changing CSM configurations. As the series of configurations cycles in and out of our perceptions of place and time, so does our sense of now.

We have given consideration to several players on the stage: the arrow of time, the second law of thermodynamics, changes, and CSM configuration. Let’s consider those players in relation to the past, present, and future: the passage of time, A-theory, B-theory, and the block universe.

2.3 The Passage of Time

We “sense” the passage of time, and many of our human languages build this concept into their structure in the form of tenses. In many ways and in many things, we sense the passage of time through motion, changing states, beginnings, and endings. Many have asked, “Is this sense that time passes all in our heads or is it real?”

Discussions of the passage of time revolve around two theories of time, known as A-theory and B-theory. Dyke notes:

There are two basic metaphysical theories of time in this dispute. There is the A-theory of time, according to which the common-sense distinction between the past, present and future reflects a real ontological distinction, and time is dynamic: what was future, is now present and will be past. Then there is the B-theory of time, according to which there is no ontological distinction between past, present and future. The fact that we draw this distinction in ordinary life is a reflection of our perspective on temporal reality, rather than a reflection of the nature of time itself. A corollary of denying that there is a distinction between past, present and future is that time is not dynamic in the way just described. The A-theory is also variously referred to as the tensed theory, or the dynamic theory of time. The B-theory is also referred to as the tenseless theory, or the static, or block universe theory of time [14].

This philosophy describes nature as indifferent to the passage of time: nature simply exists. Universal matter,

{f}, transitions to sub-f particles of matter. Configurations of universal and sub-f particle matter are replaced with new configurations. This is nature, but nature is not an entity: it has no thoughts or senses. Thus, our perceptions of the passage of time are valid; configuration changes can be viewed as “time,” but our interpretation of those changes is strictly human. Thus, this philosophy of nature aligns with B-theory (above).

2.4 The Block Universe

The philosophy, then, also aligns well with Falk, who states:

Many physicists argue that Einstein’s position is implied by the two pillars of modern physics: Einstein’s masterpiece, the general theory of relativity, and the Standard Model of particle physics. The laws that underlie these theories are time-symmetric — that is, the physics they describe is the same, regardless of whether the variable called “time” increases or decreases. Moreover, they say nothing at all about the point we call “now” — a special moment (or so it appears) for us, but seemingly undefined when we talk about the universe at large. The resulting timeless cosmos is sometimes called a “block universe” — a static block of spacetime in which any flow of time, or passage through it, must presumably be a mental construct or other illusion [15].

2.5 Measured Time

Time, then, consists of changes in CSM configurations. Our interpretation of them is strictly for our convenience. We are almost done discussing the players on the stage. Let’s discuss measured time.

According to the *Science Council*, “Science is the pursuit and application of knowledge and understanding of the natural and social world following a systematic methodology based on evidence” [16]. Advances in the measurement of time have been invaluable to the scientific method. An entry in the *Encyclopedia Britannica* states, “Although defining time presents difficulties, measuring it does not; it is the most accurately measured physical quantity” [17].

We can certainly call this type of time “real.” The reason is simple: changes over measured intervals produce changes in CSM configurations. But the relationship between time and CSM configurations is variable: a measured interval of time does not and cannot directly quantify the amount or nature of CSM configuration changes.

2.6 Existential Time

Finally, let's discuss existential time. Existential time is "the subjective, qualitative experience of time as lived and felt by a conscious human being, rather than the steady ticking of a clock" (Google AI summary). The

changes in CSM configuration that occur while an entity is "being" are very real. While we meditate, chant, or try to stop the internal dialogue, the environment surrounding our CSM continues to change. We may try to tune out time, but the transitional changes of our CSM configuration still occur all around us.

Part 6

The Expanding Universe

1 Introduction

Regardless of how we look at time, it reflects physical changes to our world. We have linked time directly to universal matter because time represents changes in CSM configurations. Let's take a brief look at the expansion of the universe and how it might relate to time. In Part 5 above, we discussed how the philosophy is related to the many aspects of time. In the discussions, there was a specific reference to the "block universe." Here the universe is viewed as tenseless, or timeless, "a static block of spacetime in which any flow of time, or passage through it, must presumably be a mental construct or other illusion" [15].

George Ellis presents an augmented view of the block universe, stating that in his model of an "evolving block universe"—

... time is a more basic entity, one that can be understood by picturing the block universe as itself evolving. In his model, the universe is a growing volume of space-time. The surface of this volume can be thought of as the present moment. The surface represents the instant where "the indefiniteness of the future changes to the definiteness of the past," as he described it. "Space-time itself is growing as time passes." One can discern the direction of time by looking at which part of the universe is fixed (the past) and which is changing (the future). ... Ellis stresses that the model is a modification, not a radical overhaul, of the standard view. "This is a block universe with dynamics covered by the general-relativity field equations — absolutely standard — but with a future boundary that is the ever-changing present,"... In this view, while the past is fixed and unchangeable, the future is open. The model "obviously represents the passing of time in

a more satisfactory way than the usual block universe," he said [15].

This is a very interesting idea. Mixing the concepts of "present" and "past" with an expanding universe is delightful. However, does the philosophy speak to an evolving block universe or to an expanding universe? Perhaps it speaks to the former and certainly to the latter.

2 The Evolving Block Universe

Although the idea of an evolving block universe is fascinating, the discussion of a surface of expansion is problematic. All that we know is inside the expansion of the universe. Is Ellis suggesting that all of the universe sits on the edge, that is, the surface of the expansion? Or is he suggesting that a "current" expansion, happening in any such location, creates a "surface wave" that carries the present along? Let's consider this second option.

We could equate an expanding block universe "surface" to an ever-changing CSM configuration. This idea grants a physical basis to the evolving block universe in the concepts of this philosophy. Can, or does, the philosophy speak to the general idea of an expanding universe?

3 The Expanding Universe

The universal join, discussed in Part 1, does present a physical basis for an expanding universe. This basis is found in one of the two aspects of time detailed in Part 5. In subsection 2.2, we combined sub-f particles of matter and the AoT under the single label "configuration." The expansion of the universe is an expression of sub-f particles of matter.

A CSM is the result of compositional processes. The byproducts of these processes are sub-f particles of matter, which form in and around the various levels of the compositional process. Sub-f particles push outward energetically in all directions.

When sub-f particles remain inside a compositional object, they provide loft such that, at any stage of the compositional process, internal build objects have access to particles in the field f . Internal maintenance is an unending need as large compositions, such as atoms, suffer the wear and tear of existence. The loft of energetic sub-f particles also keeps the atoms of a CSM from collapsing under the gripping crush of gravity.

Sub-f particles remaining independent of compositional objects provide the basis of expansion in the universe. There is an inverse relationship between the proximity of sub-f particles to a CSM and the effectiveness of their ability to expand space. Sub-f particles that are held close to a CSM have their expansion abilities mostly

negated. On the other hand, sub-f particles that are distant from CSMs are much more effective in expanding the universe.

The ability of sub-f particles to expand space is not uniform; it depends on the amount of universal join activity in the open reaches of space. Greater joining activity equates to a slight increase in expansion, and vice versa in areas with less joining activity. Differences in expansion are slight because of the overwhelming volume of space.

A 2020 study published in *Astronomy & Astrophysics* supports this idea. Lead author K. Migkas reports that the universe may not be isotropic [18]. In his article for NASA, Mohan quotes Migkas: “One of the pillars of cosmology – the study of the history and fate of the entire universe – is that the universe is ‘isotropic,’ meaning the same in all directions,” said Konstantinos Migkas of the University of Bonn in Germany, who led the new study. “Our work shows there may be cracks in that pillar” [19].

Part 7

Light

1 Preface to Light

Light holds a substantial place in the hearts and minds of people. Its importance cannot be overestimated in many scientific areas, not the least of which is physics. It is central to Albert Einstein’s theories of relativity, in which he considered the speed of light in a vacuum to be constant.

The exact nature of light is well studied, but in many ways, it remains a mystery. It is said to have zero mass, but it is affected by gravity and can affect some objects. By considering light as energy carried in electromagnetic waves, we can gain insight from this philosophy.

The final step in making an atom of hydrogen was showing that a simple composite particle (SCP) could become entwined in the conductive flow of an effective triple prime composite particle. The mechanics of quantitative joining in the effective triple prime composite particle establish the electron’s place and path in the hydrogen atom. In discussing this joining behavior, we will examine another type of advanced composite level-three particle, a complementary particle (CmP), and its role in the production of light.

2 Regular Atomic State

Under ordinary conditions, the relationship between the proton core and the SCP electron is stable. The proton pulls the SCP inward, repeatedly attempting a quantitative join with it; the SCP resists the pull because it is embedded in the proton’s conductive flow.

The pull and release of the SCP establish a wave-like pattern. Baird notes, “When bound in a stable state in an atom, an electron behaves mostly like an oscillating three-dimensional wave, i.e. the orbital vibrates. It’s a bit like a vibrating guitar string. When you pluck a guitar string, you get the string shaking, which is what creates the sound. Scientifically, we would say that you have excited a standing wave in the string” [20].

The wave oscillates between strong and weak states, a pattern that Baird also notes: “An electron in an atomic orbital state acts somewhat like a plucked guitar string. It is spread out in a three-dimensional cloud-like wavefunction that vibrates. Whereas a guitar string vibrates up and down, an atomic electron wavefunction simply vibrates strong and weak. The frequency at which the electron wavefunction vibrates is directly proportional

to the total energy of the electron” [20]. The weak part of the wave correlates with the quantitative pull on the electron because the electron is weak enough for the proton to pull it toward itself. The strong part of the wave correlates with the rebounding escape of the embedded electron.

However, if an external source of energy is applied to the atom, the relationship between the proton and the electron can change. In specific cases, an electron may jump into an orbital that is farther from the proton core, a condition known as an excited state. Causes include photon absorption, particle collisions, thermal heating, and electrical discharges. The opposite is also true: the electron of an atom can lose energy and return to its ground state. One of the ways that an electron returns to its ground state is by emitting a photon.

3 The Spring

The pull-and-release action of the proton-electron relationship exploits the electron’s conducted anchor to release a photon while the electron is in an excited state. The extra energy of the excited state is transferred to the electron’s conducted path and acts as a “spring” that ejects a photon. The photon is a CmP that is caught up in the electron’s conducted flow and is located near the electron, as illustrated in the following figures. As the proton releases its grip on the electron, the conducted flow springs out and away from the proton and the CmP is emitted as radiation. Note that the shaded layers shown in Figure 25 are detailed in Part 3.

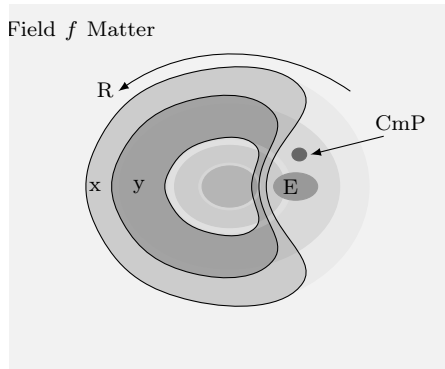


Figure 23: Spring Action Ready

The electron, labeled “E,” is drawn inward toward the core of the proton, along with the outer layers of energetic conducted matter. The compressed layers, labeled “x” and “y,” build up potential energy as they are pulled inward by the proton’s quantitative join efforts with the electron. All of the conducted layers rotate in direction “R.” The CmP is illustrated beside the electron, and it is similar to, but smaller than, the electron.

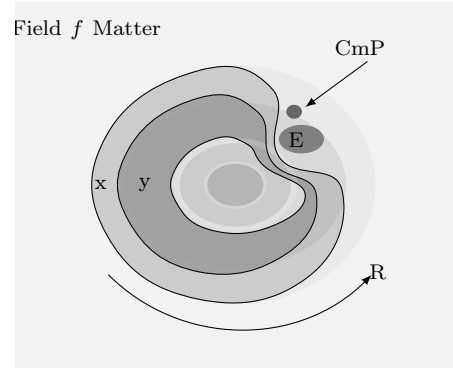


Figure 24: Spring Executing

As the proton fails in its quantitative join efforts with the electron and the electron is properly embedded in its conducted flow, the compressed layers of conducted potential energy are released and spring outward. The compressed indentation serves as a wave carrying the electron and the CmP with it.

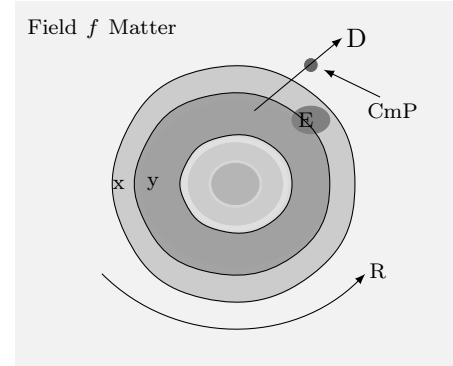


Figure 25: Light Emitted

As the spring action completes its cycle, the electron settles back into its ground state near the edge of the energetic conducted flow. The CmP, ejected outward in direction “D,” is the result of the springing wave motion described above. The springing wave imparts energy to the CmP in the form of the particle’s velocity and rotations.

4 Energy

As described in section 3, our particle of light is not yet fully formed: an expression of energy in the particle and in the field f is needed to complete the process. The CmP, our particle of light, is now moving outward in direction “D,” along the y-axis, as shown in Figure 26. As discussed in subsection 4.4, the CmP is represented by two ellipsoids composed of shaded frames. The rotations of the ejected particle are:

- End-over-end rotation on the y- and z-axes, with the emitter end marked by a red dot; and

- Around-and-around rotation on the x- and y-axes, with the emitter end marked by a blue dot.

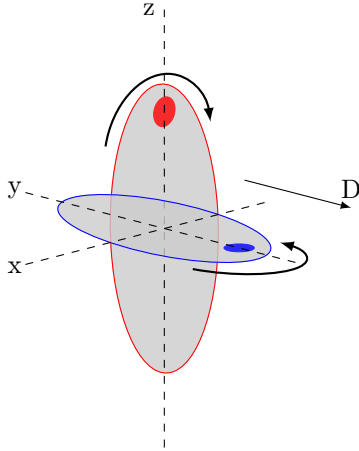


Figure 26: Complementary Particle Rotations

Locally, in the immediate area of the atom, particles are successfully emitted. Nonlocally, however, ejected particles may or may not continue to propagate as light as their distance from the atom increases. Successful nonlocal propagation is determined by the amount of energy imparted to the particle when it is ejected.

4.1 Energy Propagation

As noted in Part 3, energy moves through the field f as a wave of rotations. The wave might include a particle embedded in one of the rotations, and such is the case for light. The springing wave that ejected the particle of light away from the atom is very energetic due to the quantitative join action's very strong pull and the excited state of the electron. That pull is released and imparts energy to the electron and the ejected particle. For the ejected CmP, that energy is equally applied to all three axes of motion. Let's consider this application of energy in terms of the particle and the field f .

4.2 Movement Along Axes

For the moment, consider energy propelling the particle along the z-axis in the direction of movement "D," as noted in Figure 27, and spinning up rotations of energy ahead of the particle. The rotations of energy in the illustration are arbitrary: in general, the number of energy rotations is proportional to the amount of energy transferred to the particle as it was ejected from the atom.

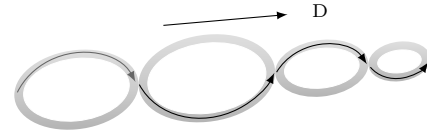


Figure 27: Energy Rotations

Let's first translate the illustration in Figure 27 to Figure 28. This is accomplished by mapping the rotations in Figure 27 onto the red-dot y-axis rotation in Figure 26. The y-axis rotations are shown as a series of frames, with the first frame marking the emitter end of the CmP with a red dot on the positive side (sine) of the rotation. The second rotation frame shows the emitter end of the CmP on the negative side (cosine) of the rotation. The illustration contains six rotation frames, with each complete sine/cosine wave cycle requiring two frames.

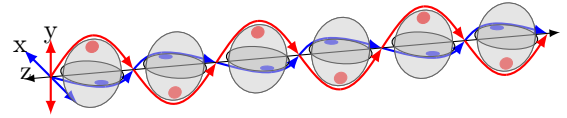


Figure 28: Energy Rotations on the x- and y-Axes

The second series, shown in Figure 26 but missing from Figure 27, is illustrated as rotations with blue highlights on the x-axis of Figure 28. As with the y-axis, each emitter/receiver rotation frame set is shown as an ellipse, with the emitter side marked by a blue dot.

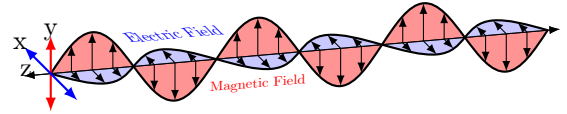


Figure 29: Classic EM Wave

Figure 29 redraws the energy rotations in Figure 28 in a form commonly found in many texts. For example, NOAA's website states, "Electromagnetic waves are a form of radiation that travel though [sic] the universe. They are formed when an electric field ...couples with a magnetic field..." [21].

4.3 Axes Energy Rotations

We now have the particle of light moving along the z-axis as a series of rotations on the x- and y-axes. We must carefully note that the series shown in Figure 28 is a composition of frames superimposed on one another to show the prolonged movement of light. However, at any such moment in "time," the most accurate depiction is the series shown in Figure 27, moving in direction "D"

along the z-axis, with rotations on the y-axis and additional rotations for the x-axis.

Let's clarify the movement and energy expressions of the CmP. While the particle is moving along the z-axis, energy is expressed as rotations on the x- and y-axes. The total energy provided to the CmP by the spring action of the parent atom is recorded in a sort of financial "ledger." The spring action initializes the ledger with a credit of energy. The CmP then goes about spinning up particles of $\{f\}$ as rotations on the x- and y-axes and, as it does so, each instance of energy use writes a debit to the ledger. The ledger balance must remain positive for the particle of light to reach the critical threshold successfully.

The CmP's spin-up rotations of $\{f\}$ matter use quantitative joins and, in so doing, increase its effective mass. At some point, the effective mass of the rotations might exceed the mass of the CmP. In this case, the energy rotations will reverse roles with the CmP and make a quantitative join with the CmP rotations.

In the philosophy, we see how a CmP becomes fixed into the conducted flow of an effective triple prime (a

proton) and assumes the role of an electron. Here, the same process takes place, but the role of the proton is now played by the two energy rotations on the x- and y-axes. In the role of a pseudo electron, the CmP conducts the rotating $\{f\}$ matter through its receiver/emitter channels. As the energy rotations absorb the CmP, any excess energy is absorbed back into the field f . Keep in mind that there are no guarantees: The energy given to the CmP by the parent atom may be insufficient for its energy rotations to exceed the mass of the CmP. In this case, the particle of light will gradually succumb to resistance and be absorbed back into the field f .

If the parent atom provides sufficient energy, the particle of light is successfully formed. The rotations of energy, together with the integrated CmP, become part of spacetime. The rotating matter does not move through space; rather, it only acts as a conductor, allowing the energy to travel through space. Thus, the CmP is integrated with the energy. Any sub- f particles of $\{f\}$ created by the process or by the particle's movement in the field remain within the $\{f\}$ matter of the moment. The particle of light, moving without its created sub- f particles, is effectively without mass and timeless.

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